

Questions with Answer Keys

MathonGo

Q1 (20 July 2021 Shift 1)

The normal reaction 'N' for a vehicle of 800 kg mass, negotiating a turn on a 30° banked road at maximum possible speed without skidding is $____ \times 10^3 \text{ kg m/s}^2$

(1) 10.2

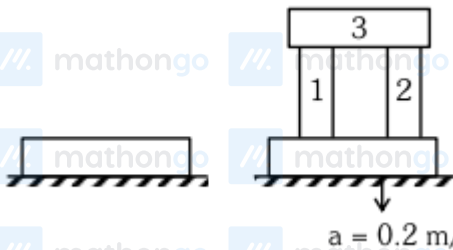
(2) 7.2

(3) 12.4

(4) 6.96

Q2 (20 July 2021 Shift 1)

A steel block of 10 kg rests on a horizontal floor as shown. When three iron cylinders are placed on it as shown, the block and cylinders go down with an acceleration 0.2 m/s^2 . The normal reaction R' by the floor if mass of the iron cylinders are equal and of 20 kg each, is N. [Take $g = 10 \text{ m/s}^2$ and $\mu_s = 0.2$]



(1) 716

(2) 686

(3) 714

(4) 684

Q3 (20 July 2021 Shift 2)

A body of mass ' m' ' is launched up on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of friction between the body and plane is $\frac{\sqrt{x}}{5}$ if the time of ascent is half of the time of descent. The value of x is

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Q4 (22 July 2021 Shift 1)

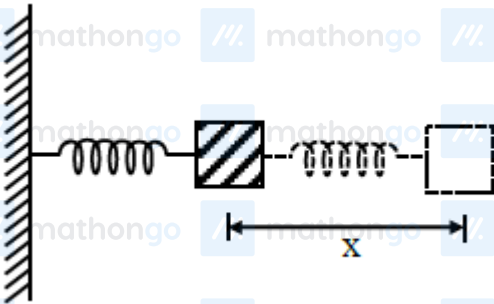
A bullet of '4g' mass is fired from a gun of mass

4 kg. If the bullet moves with the muzzle speed of 50 ms^{-1} , the impulse imparted to the gun and velocity of recoil of gun are :

- (1) 0.4 kg ms^{-1} , 0.1 ms^{-1}
- (2) 0.2 kg ms^{-1} , 0.05 ms^{-1}
- (3) 0.2 kg ms^{-1} , 0.1 ms^{-1}
- (4) 0.4 kg ms^{-1} , 0.05 ms^{-1}

Q5 (22 July 2021 Shift 1)

The motion of a mass on a spring, with spring constant K is as shown in figure.



The equation of motion is given by $x(t) = A \sin \omega t + B \cos \omega t$ with $\omega = \sqrt{\frac{K}{m}}$

Suppose that at time $t = 0$, the position of mass is

$x(0)$ and velocity $v(0)$, then its displacement can

also be represented as $x(t) = C \cos(\omega t - \phi)$, where C

and ϕ are:

- (1) $C = \sqrt{\frac{2v(0)^2}{\omega^2} + x(0)^2}$, $\phi = \tan^{-1} \left(\frac{v(0)}{x(0)\omega} \right)$
- (2) $C = \sqrt{\frac{2v(0)^2}{\omega^2} + x(0)^2}$, $\phi = \tan^{-1} \left(\frac{x(0)\omega}{2v(0)} \right)$
- (3) $C = \sqrt{\frac{v(0)^2}{\omega^2} + x(0)^2}$, $\phi = \tan^{-1} \left(\frac{x(0)\omega}{v(0)} \right)$
- (4) $C = \sqrt{\frac{v(0)^2}{\omega^2} + x(0)^2}$, $\phi = \tan^{-1} \left(\frac{v(0)}{x(0)\omega} \right)$

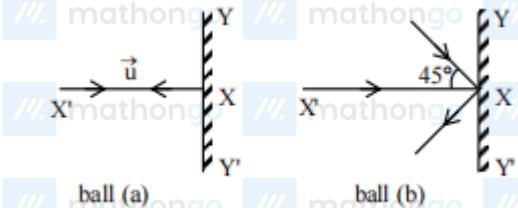
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Q6 (25 July 2021 Shift 1)

Two billiard balls of equal mass 30 g strike a rigid wall with same speed of 108 kmph (as shown) but at different angles. If the balls get reflected with

the same speed then the ratio of the magnitude of impulses imparted to ball 'a' and ball 'b' by the wall along 'X' direction is :



- (1) 1 : 1
- (2) $\sqrt{2} : 1$
- (3) 2 : 1
- (4) $1 : \sqrt{2}$

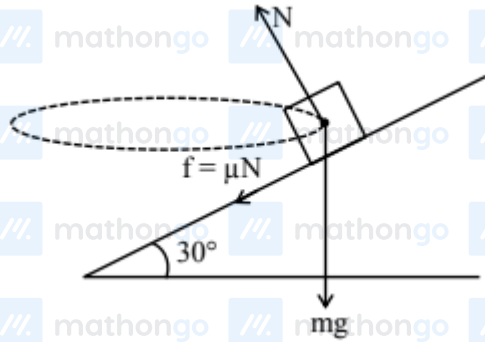
Q7 (25 July 2021 Shift 2)

A force $\vec{F} = (40\hat{i} + 10\hat{j})\text{N}$ acts on a body of mass 5 kg. If the body starts from rest, its position vector $\vec{r}$ at time $t = 10\text{ s}$, will be :

- (1) $(100\hat{i} + 400\hat{j})\text{m}$
- (2) $(100\hat{i} + 100\hat{j})\text{m}$
- (3) $(400\hat{i} + 100\hat{j})\text{m}$
- (4) $(400\hat{i} + 400\hat{j})\text{m}$

Answer Key**Q1 (1)****Q2 (2)****Q3 (3)****Q4 (2)****Q5 (4)****Q6 (2)****Q7 (3)**

Q1



At $v_{\max}$, f will be limiting in nature.

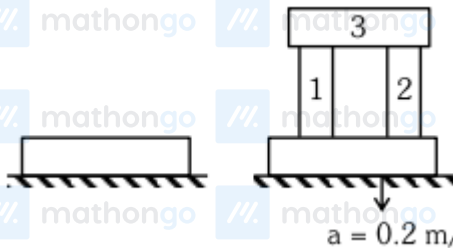
$\therefore$ Balancing force in vertical direction, $N \cos 30^\circ - mg - \mu N \cos 60^\circ = 0$

$$\Rightarrow N [\cos 30^\circ - \mu \cos 60^\circ] = mg$$

$$\therefore N = \frac{800 \times 10}{(0.87 - 0.1)} \approx 10.2 \times 10^3 \text{ kg m/s}^2$$

Hence option 1.

Q2



Writing force equation in vertical direction

$$Mg - N = Ma$$

$$\Rightarrow 70g - N = 70 \times 0.2$$

$$\Rightarrow N = 70[g - 0.2] = 70 \times 9.8$$

$$\therefore N = 686 \text{ Newton}$$

Note : Since there is no compressive normal from

the sides, hence friction will not act.

Hence option 2.

Hints and Solutions

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Q3

$$t_a = \frac{1}{2} t_d$$

$$a \propto \sqrt{\frac{2s}{a_a}} = \frac{1}{2} \sqrt{\frac{2s}{a_d}} \dots (i)$$

$$a_a = g \sin \theta + \mu g \cos \theta$$

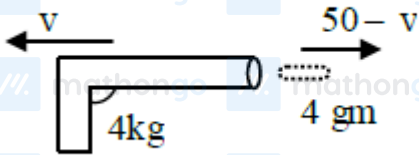
$$= \frac{g}{2} + \frac{\sqrt{3}}{2} \mu g$$

$$a_d = g \sin \theta - \mu g \cos \theta$$

$$= \frac{g}{2} - \frac{\sqrt{3}}{2} \mu g$$

using the above values of a_a and a_d and putting in (i) we will get $\mu = \frac{\sqrt{3}}{5}$ equation

Q4



By momentum conservation

$$4 \times 10^{-3} (50 - v) - 4v = 0$$

$$v = \frac{4 \times 10^{-3} \times 50}{4 + 4 \times 10^{-3}} \approx 0.05 \text{ ms}^{-1}$$

$$\text{Impulse } J = mv = 4 \times 0.05 = 0.2 \text{ kgms}^{-1}$$

Q5

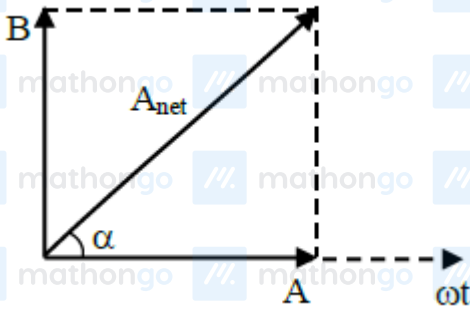
$$x = A \sin \omega t + B \cos \omega t$$

$$v = \frac{dx}{dt} = A \omega \cos \omega t - B \omega \sin \omega t$$

$$\text{At } t = 0, x(0) = B$$

$$v(0) = A \omega$$

$$x = A \sin \omega t + B \sin(\omega t + 90^\circ)$$



$$A_{\text{net}} = \sqrt{A^2 + B^2}$$

$$\tan \alpha = \frac{B}{A} \Rightarrow \cot \alpha = \frac{A}{B}$$

$$\Rightarrow x = \sqrt{A^2 + B^2} \sin(\omega t + \alpha)$$

$$\Rightarrow x = \sqrt{A^2 + B^2} \cos(\omega t - (90 - \alpha))$$

$$x = C \cos(\omega t - \phi)$$

$$\Rightarrow C = \sqrt{A^2 + B^2}$$

$$C = \sqrt{\frac{[v(0)]^2}{\omega^2} + [x(0)]^2}$$

$$\phi = 90 - \alpha$$

$$\tan \alpha = \cot \alpha = \frac{A}{B}$$

$$\Rightarrow \tan \phi = \frac{v(0)}{x(0) \cdot \omega}$$

$$\phi = \tan^{-1} \left(\frac{v(0)}{x(0) \omega} \right)$$

Q6

Impulse = change in momentum

$$\text{Ball (a)} \quad |\Delta p| = 2mu = J_1$$

$$\text{Ball (b)} \quad |\Delta p| = 2mu \cos 45^\circ = J_2$$

$$\frac{J_1}{J_2} = \frac{1}{\cos 45^\circ} = \sqrt{2}$$

Q7

$$\frac{d\vec{v}}{dt} = \vec{a} = \frac{\vec{F}}{m} = (8\hat{i} + 2\hat{j})\text{m/s}^2$$

$$\frac{d\vec{r}}{dt} = \vec{v} = (8t\hat{i} + 2t\hat{j})\text{m/s}$$

$$\vec{r} = (8\hat{i} + 2\hat{j}) \frac{t^2}{2} \text{m}$$

Hints and Solutions

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At $t = 10\text{sec}$

$$\vec{r} = [(8\hat{i} + 2\hat{j})50]\text{m}$$

$$\Rightarrow \vec{r} = (400\hat{i} + 100\hat{j})\text{m}$$